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Education

FAST - National University of Computer & Emerging Sciences

Lahore, PK

BS Data Science

2023-2027

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Systems, Software Engineering

Experience

Speedy Drive Car Rental, Dubai, UAE

Remote

AI Engineer (Voice AI and Automation)

Sept 2025 – Feb 2026

- Deployed a production **Voice AI Agent** handling end-to-end car rental sales calls — availability, dynamic pricing, insurance, delivery, and booking confirmation — with **zero human intervention**
- Engineered real-time **backend tool functions** for live availability lookup, **SQL**-based pricing, total cost aggregation, and **TTS-formatted** automated booking confirmation emails
- Implemented **Vapi** tool-calling architecture for mid-call structured data injection, eliminating pricing and availability hallucinations via strict tool orchestration
- Developed a real-time **analytics dashboard** (TypeScript) tracking call outcomes, booking statuses, success rates, and revenue across Daily/Weekly/Monthly views
- Engineered **agent system prompt** covering multi-department routing, dynamic tool orchestration, fallback handling, TTS-compliant formatting, and conditional booking logic

Projects

AI Virtual Try-On

TypeScript, Fastify, BullMQ, Redis, REST APIs

- Architected an **async backend pipeline** using Fastify with multipart endpoints accepting product image URLs and user-uploaded images for AI try-on processing
- Implemented **BullMQ + Redis job queue** with full lifecycle tracking (queued → processing → completed) and real-time status polling via REST
- Added UUID-based file storage, input validation, and structured error handling for reliable async task processing at scale

Business Insights and Sales Forecasting Tool

Python, FastAPI, Node.js, PostgreSQL, Prophet, Random Forest, TensorFlow/Keras, LangGraph, LangChain, ChromaDB, React

- Full-stack **decision intelligence platform** with a Node.js/PostgreSQL operational backend for inventory, stock movement, PO processing, and supplier management, and a Python FastAPI analytical service hosting ML models and AI agents
- Time-series **sales forecasting** pipeline using Prophet (baseline) and Random Forest with iterative lag-feature inference (production); predictions served via **/api/forecast** with MAE/RMSE/MAPE monitoring
- Customer **churn prediction** via TensorFlow DNN (RFM features, SMOTE oversampling, AUC-ROC ≈0.82) with Risk segmentation and live benchmarking against Logistic Regression, Random Forest, and KNN baselines

- Multi-agent **AI consultant chatbot** using LangGraph (Router → Planner → Data Gatherer → RAG Retriever → Synthesizer) with Llama 3.1/3.3, ChromaDB vector store, and live SQL + ML tool calls

Smart Movie Recommendation System

Python, NLP, scikit-learn, EDA, TMDB API

2025

- Performed **EDA** analyzing genre distributions, popularity trends, and feature correlations to guide model design
- NLP-based **content-based recommendation engine** consolidating genres, cast, crew, and keywords into a unified tags vector for cosine similarity matching
- Integrated **TMDB API** to render posters and metadata for top-5 recommendations in an interactive UI

Airline Customer Satisfaction Predictor

Python, scikit-learn, Random Forest, Logistic Regression, KNN, EDA

- Conducted **EDA** identifying key satisfaction drivers: seat comfort, service quality, flight distance, and customer loyalty patterns
- Multi-model **binary classification pipeline**; Random Forest achieved best accuracy, precision, recall, and F1 over Logistic Regression and KNN baselines

Skills & Interests

Voice AI and Agents: Voice AI Agent Development, Tool-Calling Architecture, Prompt Engineering, Agent System Design, TTS Optimization, LLM Orchestration, Workflow Automation

Machine Learning and AI: scikit-learn, XGBoost, Random Forest, NLP, Feature Engineering, Content-Based Filtering, TimeSeries Forecasting, EDA, Data Visualization

Backend and APIs: TypeScript, Node.js, Fastify, REST APIs, BullMQ, Redis, Async Pipelines, Multipart Handling, Third-Party API Integration

Data and Databases: Python, SQL (relational schema design, lead storage, CRM pipelines), Pandas, Data Cleaning and Preprocessing

Tools: Git, GitHub, Postman, n8n, make, Retell, PowerBI,